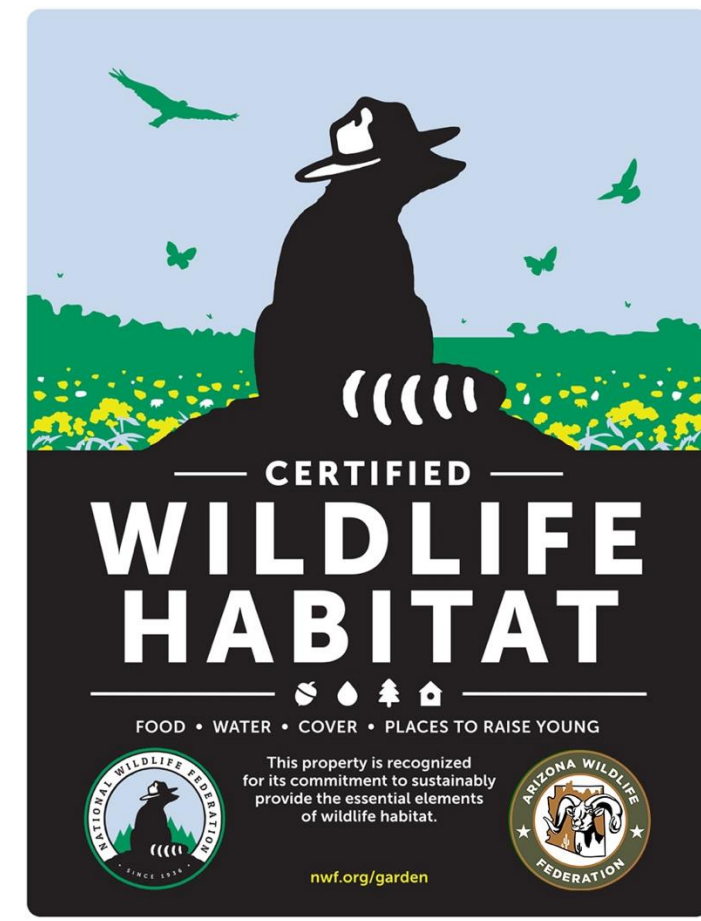




Geospatial and Temporal Trends of Wildlife Gardening in Arizona & Metro Phoenix



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INTRODUCTION

Past research has shown that gardens in urban areas can significantly increase vegetation and bird biodiversity while also supporting other wildlife such as pollinators (Lerman et al. 2021). To understand where people are most motivated to garden for wildlife in Arizona and metro Phoenix, our team geolocated Arizona properties where people have participated in the **National Wildlife Federation's (NWF) Certified Wildlife Habitat © program**, which motivates and provides resources for people to manage landscapes to support wildlife such as birds and pollinators. Partnering with NWF and the state chapter, the **Arizona Wildlife Federation (AWF)**, we examined the spatial and temporal trends in the state and metro Phoenix area, along with the property types certified since 1972.

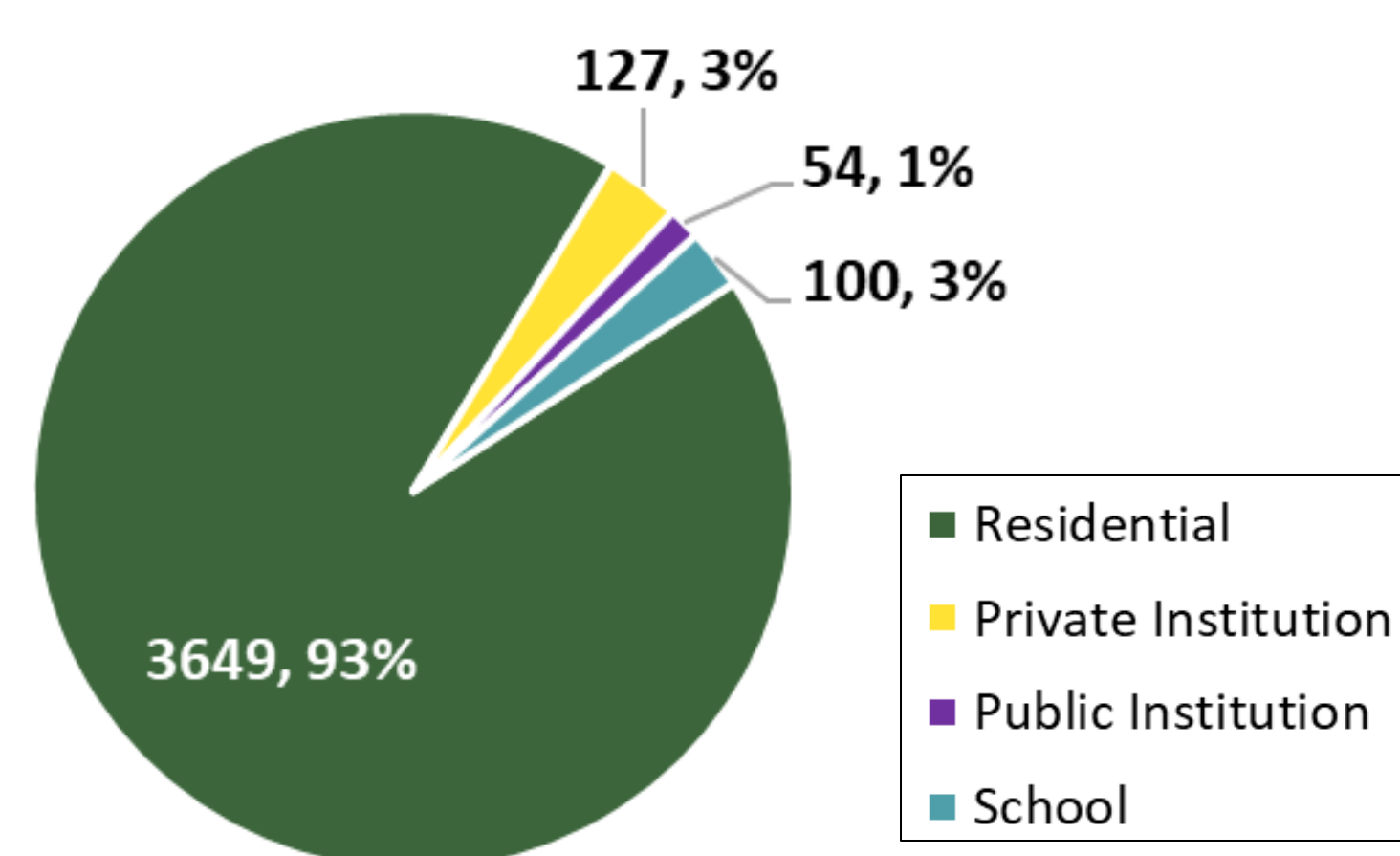
Research Question: *Where do people garden for wildlife in Arizona and metro Phoenix; how and why do NWF-certified habitats vary spatially and temporally?*

METHODS

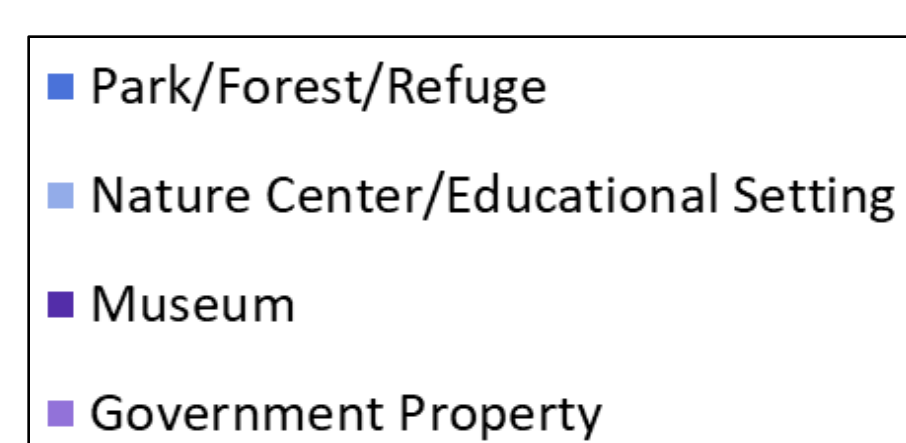
For the NWF certification program, gardening for wildlife involves managing land as habitat by providing food, water, cover, and places to rear young, in addition to using sustainable management practices (e.g., no chemicals). The NWF data analyzed herein represent geolocatable habitats certified on different types of properties between 1972-2024 in the state of Arizona. Our team used Microsoft Excel to clean the data before projecting it in ArcGIS Pro using USA Contiguous Albers Equal Area Conic projection. We then evaluated 1) *the types of properties certified (public or private)*, 2) *the number of certifications by year*, and 3) *their location*. We met with the AWF to discuss the analysis, map design, and draft blogs roughly every two weeks. **For more on our team's AWF blog posts, see the poster by Lexi Cegielski and Sharika Kapur.**

PROPERTY TYPES

Certified Wildlife Habitat by Property Type (3930)

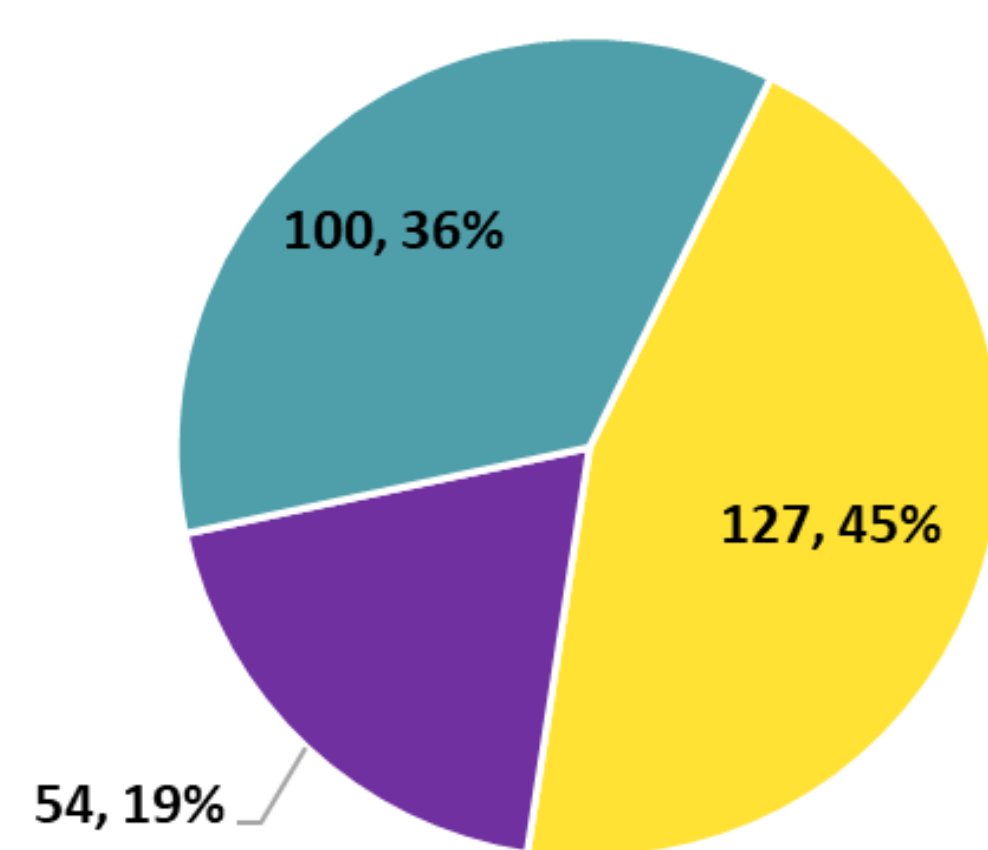


Of the *non-residential habitats*, most of them are K-12 schools (93) and farms (64), in addition to businesses and corporations (49). The purple color represents entities in the public sector including:



Residential properties consist of 93% of the NWF-certified habitats in AZ, most of which are yards with a smaller number (n=40) of community gardens. Schools, public institutions, and private institutions host the remaining 7% of habitats, as seen below.

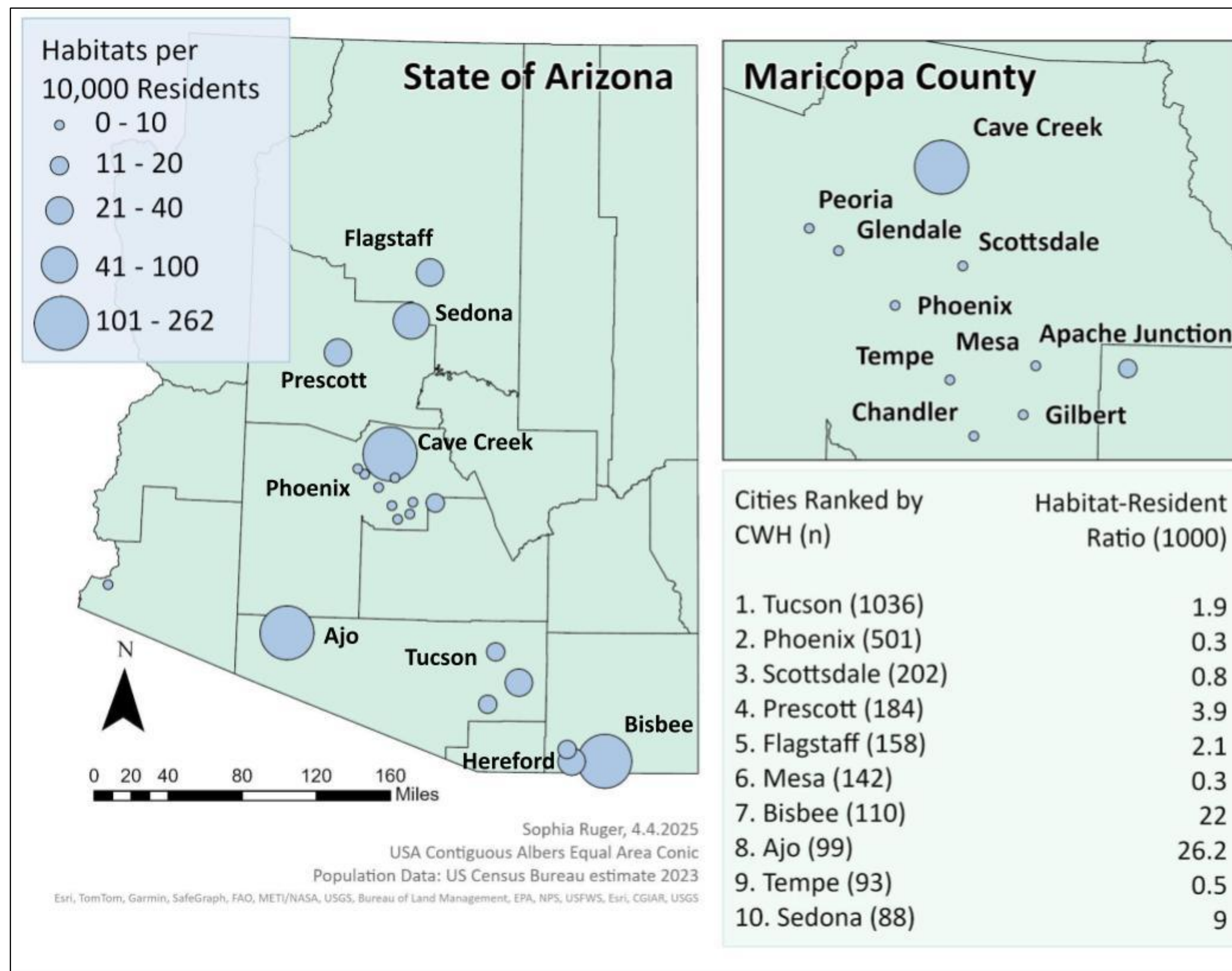
Non-Residential Certified Wildlife Habitat (281)



SPATIAL DISTRIBUTION

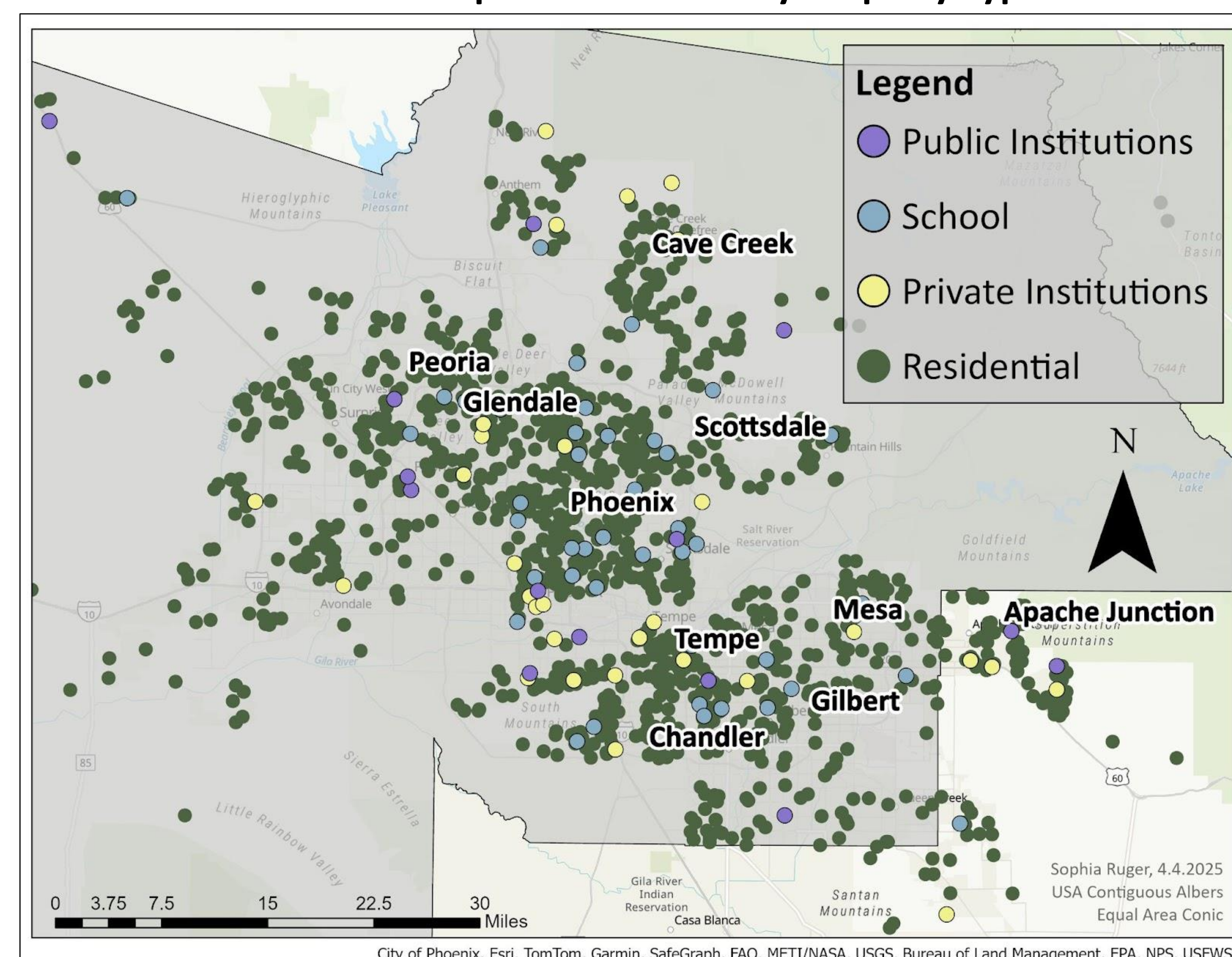
The map below symbolizes the amount of NWF-certified habitats per 10,000 residents for the top 20 towns or cities in Arizona. Although larger cities including Tucson and Phoenix have more certified habitats, smaller towns like Ajo and Bisbee have more gardens per capita. Towns such as Sedona, Prescott, and Flagstaff also have relatively large numbers of certified habitats. Zooming into the Phoenix area, Cave Creek and Apache Junction have the highest number of gardens per capita.

Certifications Normalized by Population of the Top 21 Municipalities



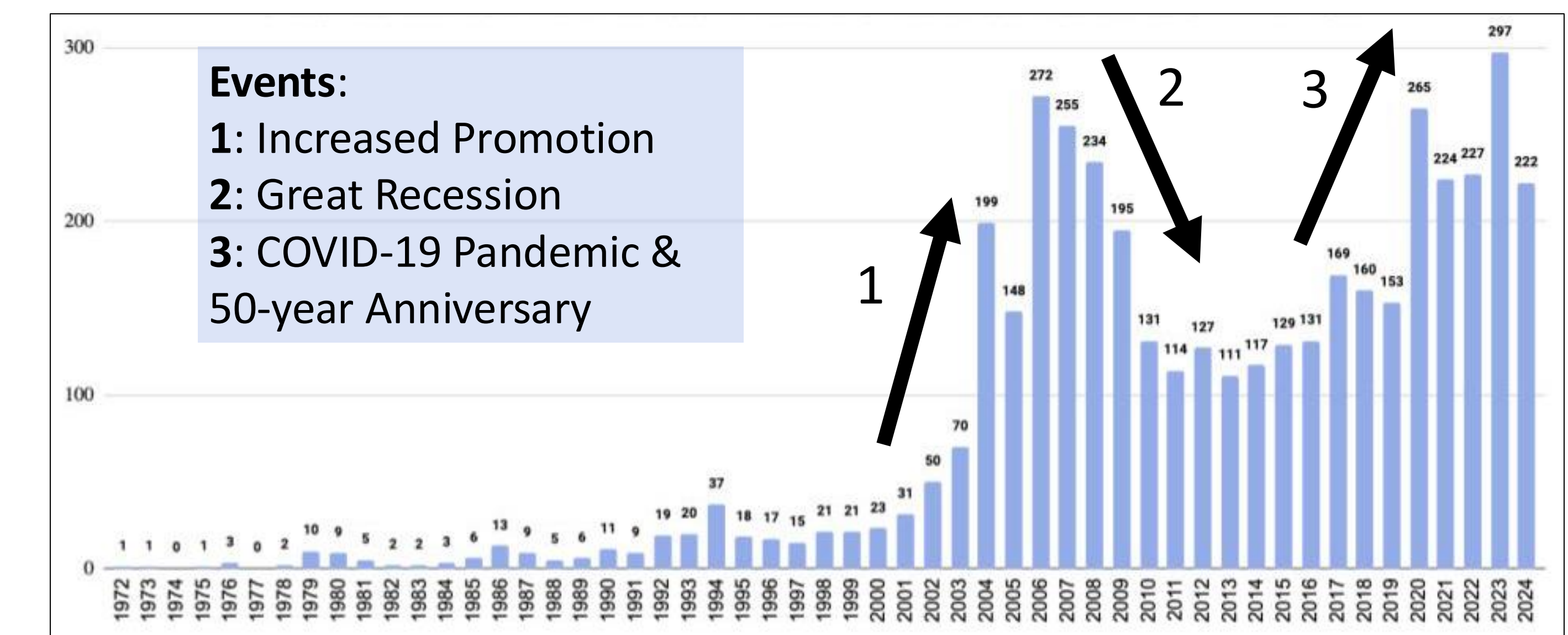
Although residential yards are the most common property type certified as habitat in metro Phoenix, K-12 schools and private institutions (including businesses) throughout the region are also relatively common (as seen in the below map).

Phoenix Metropolitan Gardens by Property Type



TEMPORAL TRENDS

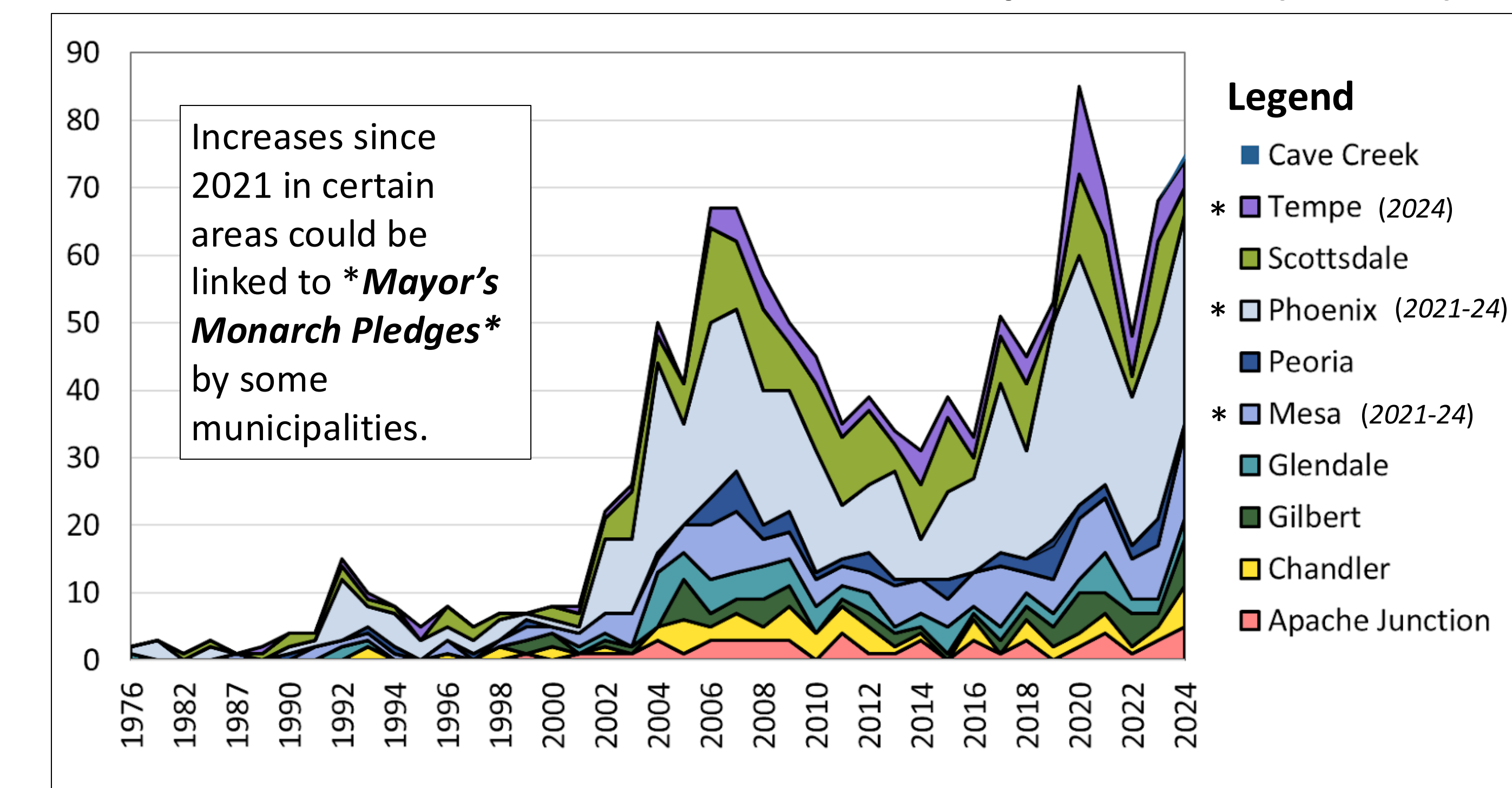
Annual Count of Arizona Certified Gardens by Year (n=4335)



Arizona (above): few properties were certified by NWF until the turn of the century, perhaps because access to the internet emerged and the NWF increasingly promoted the program post-2000 (1), when certifications grew to a high of 272 in 2006. In the recession years (2), certifications dropped to a low of 111 in 2014. Around the pandemic, certifications again rose (3). Around the program's 50-year anniversary, certifications reached a new high of 297 gardens in 2023.

Phoenix (below): the metro region mimics the state-wide trends, with Phoenix proper leading certifications. Scottsdale, Tempe, and Phoenix share very similar trends, whereas Peoria, Mesa, and Glendale share peaks and valleys in the same years. The most certifications in the metro area were in 2020, during the pandemic, compared to the 2023 NWF anniversary for that state.

Garden Certifications in the Phoenix Metropolitan Area (n=1230)



DISCUSSION

As our work continues, we will delve into why certain communities see more certifications than others based on top-down efforts (e.g., NWF and AWF programming and municipal actions) and bottom-up efforts (e.g., local actors seeking community certification). We will also explore how and why K-12 schools get certified in Maricopa County, which has about half of NWF-certified habitats in Arizona (n=46 of 93). Schools have the potential to inspire environmental stewardship in the youngest generation while protecting wildlife and enhancing hands-on, experiential learning. Therefore, our research will assess where and why schools participate and the associated outcomes.

Acknowledgements: Thank you to the National and Arizona Wildlife Federation for supporting this research, particularly Mary Phillips, Keith Ashley, and Elise Lange. This material is based upon work supported by the National Science Foundation under grant DEB-2224662, Central Arizona-Phoenix Long-Term Ecological Research Program (CAP LTER). The citation included is: Lerman, Narango et al. 2021. Residential yard management and landscape cover affect urban bird community diversity across the continental USA. *Ecological Applications*, 31(8), p.e02455.